(a) Show that if you cut A into B and C by a straight cut, then $D(A) = D(B) + D(C)$.

(b) Show that $\alpha = \arccos(1/3)/\pi$ is not a rational number.

Hint: Assume that $\alpha = 2m/n$, for integers m, n. Deduce that roots of the equation $x + x^{-1} = 2/3$ are roots of unity of degree n. Then show that $x^k + x^{-k}$ has denominator 3^k and get a contradiction.

(c) Using (a) and (b), show that the answer to Hilbert's question is negative. (Compute the Dehn invariant of the regular tetrahedron and the cube.)

2.14. Tensor products and duals of representations of Lie algebras

Definition 2.14.1. The **tensor product** of two representations V, W of a Lie algebra $\mathfrak{g}$ is the space $V \otimes W$ with

$$\rho_{V\otimes W}(x) = \rho_V(x) \otimes \mathrm{Id} + \mathrm{Id} \otimes \rho_W(x).$$

Definition 2.14.2. The **dual representation** V^* to a representation V of a Lie algebra $\mathfrak{g}$ is the dual space V^* to V with $\rho_{V^*}(x) = -\rho_V(x)^*$.

It is easy to check that these are indeed representations.

Problem 2.14.3. Let V, W, U be finite dimensional representations of a Lie algebra $\mathfrak{g}$. Show that the space $\mathrm{Hom}_{\mathfrak{g}}(V \otimes W, U)$ is isomorphic to $\mathrm{Hom}_{\mathfrak{g}}(V, U \otimes W^*)$. (Here $\mathrm{Hom}_{\mathfrak{g}} := \mathrm{Hom}_{\mathcal{U}(\mathfrak{g})}$.)

2.15. Representations of $\mathfrak{sl}(2)$

This subsection is devoted to the representation theory of $\mathfrak{sl}(2)$, which is of central importance in many areas of mathematics. It is useful to study this topic by solving the following sequence of exercises, which every mathematician should do, in one form or another.

Problem 2.15.1. According to the above, a representation of $\mathfrak{sl}(2)$ is just a vector space V with a triple of operators E, F, H such that $HE - EH = 2E,\ HF - FH = -2F,\ EF - FE = H$ (the corresponding map ρ is given by $\rho(e) = E, \rho(f) = F,\ \rho(h) = H$).